

SACRAMENTO MATHER, CA, USA

WMO: 724833

Lat: 38.567N

Lon: 121.300W

Elev: 99

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 86-19

WBAN: 23206

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.2	30.9	17.8	13.5	45.0	22.2	16.7	45.3	29.7	52.1	26.7	51.0	2.4	110	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.2	101.2	68.7	98.3	67.4	94.8	66.5	70.9	96.6	69.3	93.6	68.0	91.3	5.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	81.5	75.7	60.0	77.5	75.9	58.6	73.7	74.6	34.5	96.9	33.3	93.9	32.2	91.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	17.0	13.8	DB	24.3	107.7	4.3	2.8	21.2	109.7	18.7	111.3	16.3	112.9	13.2	114.9
			WB	22.6	74.4	3.8	1.5	19.9	75.5	17.7	76.4	15.6	77.2	12.8	78.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	45.4	50.3	54.4	58.1	64.2	70.7	76.4	74.6	71.2	64.3	52.5	45.2	(6)
	DBStd	12.16	4.92	5.43	5.10	6.22	7.36	6.39	5.19	5.23	5.70	6.14	5.25	5.60	(7)
	HDD50	448	159	59	17	9	4	0	0	0	0	1	34	165	(8)
	HDD65	2774	608	410	329	219	102	20	0	1	12	85	376	612	(9)
	CDD50	4345	16	69	154	253	444	623	818	763	636	444	108	17	(10)
	CDD65	1197	0	0	0	13	78	192	353	299	199	63	0	0	(11)
	CDH74	17082	0	1	48	373	1438	2886	4707	3919	2671	1026	13	0	(12)
	CDH80	9098	0	0	4	116	632	1498	2814	2239	1400	395	0	0	(13)
(14) Wind	WSAvg	5.3	4.2	5.2	6.0	5.8	5.9	6.3	6.0	5.6	4.8	4.3	4.3	5.4	(14)
(15) Precipitation	PrecAvg	16.5	3.3	2.9	2.5	1.1	0.4	0.2	0.0	0.1	0.3	0.9	2.3	2.4	(15)
	PrecMax	33.5	9.1	8.6	7.1	4.2	2.1	1.3	0.8	0.6	2.8	6.9	7.4	6.3	(16)
	PrecMin	6.3	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.3	2.5	2.3	1.9	1.1	0.5	0.2	0.2	0.2	0.6	1.2	2.0	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	71.7	79.2	88.4	95.3	101.8	106.9	104.4	100.7	94.8	75.7	63.9	(19)
		MCWB	53.3	56.6	61.6	64.0	66.6	67.9	71.4	68.9	68.3	65.0	59.0	53.3	(20)
	2%	DB	59.6	68.2	74.3	83.6	91.3	97.4	101.7	100.1	96.6	89.2	71.4	60.6	(21)
		MCWB	51.1	56.0	59.5	61.6	65.6	67.1	69.8	67.5	65.5	62.4	57.2	51.4	(22)
	5%	DB	56.8	64.3	70.3	78.0	87.9	93.2	98.6	96.7	92.6	84.0	67.9	57.2	(23)
		MCWB	51.1	54.1	57.6	59.2	64.3	65.7	68.4	66.7	64.4	61.1	54.9	50.8	(24)
	10%	DB	54.3	61.0	66.2	73.1	82.4	89.3	94.4	92.2	89.1	79.9	64.0	54.6	(25)
		MCWB	49.5	52.3	56.2	57.6	61.6	64.7	67.2	65.8	63.5	59.9	53.8	49.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	60.1	64.0	67.7	69.4	71.5	73.3	71.8	70.7	66.2	62.2	57.4	(27)
		MCDB	60.0	67.7	75.6	84.1	89.6	96.7	101.4	99.1	96.2	88.8	72.1	61.2	(28)
	2%	WB	54.5	57.3	60.8	63.4	67.8	69.4	71.3	69.6	67.9	64.1	59.1	54.3	(29)
		MCDB	57.2	64.4	71.2	78.7	88.2	92.1	98.1	95.1	91.0	84.8	66.1	56.4	(30)
	5%	WB	52.2	55.4	59.1	60.9	65.5	67.6	69.5	68.2	65.7	62.6	57.1	52.5	(31)
		MCDB	55.8	61.7	68.5	74.8	84.1	89.6	95.1	92.6	88.1	80.9	63.9	55.5	(32)
	10%	WB	50.2	53.9	57.3	58.8	63.0	65.9	68.1	66.7	64.2	61.1	55.4	50.4	(33)
		MCDB	54.1	59.5	64.7	71.1	79.3	86.5	92.2	89.6	85.8	76.7	61.4	54.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	20.0	21.9	26.8	31.3	33.6	35.2	34.4	32.2	28.1	21.8	16.4	(35)
		MCDBR	20.3	24.5	30.4	35.5	38.1	40.9	38.7	39.3	37.8	35.5	28.0	20.8	(36)
	5% WB	MCWBR	13.8	14.2	17.1	16.9	15.8	16.5	13.8	13.6	14.2	14.7	15.0	13.2	(37)
		MCWBR	17.3	20.5	25.4	31.7	35.5	37.5	37.1	36.3	35.0	31.4	22.9	16.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.326	0.347	0.356	0.369	0.364	0.364	0.348	0.355	0.351	0.333	0.336	(40)
		taud	2.501	2.504	2.430	2.402	2.384	2.409	2.410	2.445	2.431	2.443	2.477	2.454	(41)
	Ebn at Noon		271	286	290	292	289	289	288	291	283	272	264	256	(42)
		Edh at Noon	25	28	33	36	38	37	37	35	33	30	25	24	(43)
(44) All-Sky Solar Radiation	RadAvg		694	1017	1450	1919	2336	2622	2609	2340	1895	1340	859	619	(44)
	RadStd		126	133	157	169	148	108	78	74	67	89	62	103	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.72	N/A	-1.94	N/A	N/A	N/A	N/A	-209	+248	+76

Nomenclature: See separate page